

**الإجابات على "كراس الإجابة"**

**Treat the following questions:**

**I- Choose the correct answer: (1pt/ correct answer)**

1- Fragmentation is the process of breaking down a database into a set of sub-databases.

- a- True
- b- False

3- Transparency for the user and the autonomy of each site are two of the objectives of distributed DBMS.

- a- True
- b- False

5- The operator of the reconstruction of the Horizontal fragmentation is the Join ( $\bowtie$ ).

- a- True
- b- False

7- Distribution of values (Mixed fragmentation): is the combination of the two previous fragmentations, horizontal and vertical.

- a- True
- b- False

9- Which of the following RDBMS is qualified as object oriented?

- a- INFORMIX
- b- ORACLE
- c- O2

2- The rules of fragmentation are: completeness, union and reconstruction.

- a- True
- b- False

4- The operator of the Horizontal fragmentation is the selection ( $\sigma$ ):

- a- True
- b- False

6- The operator of the reconstruction of the distribution of attributes (vertical fragmentation) is the Semi-Join ( $\ltimes$ ).

- a- True
- b- False

8- Prolog is a programming language:

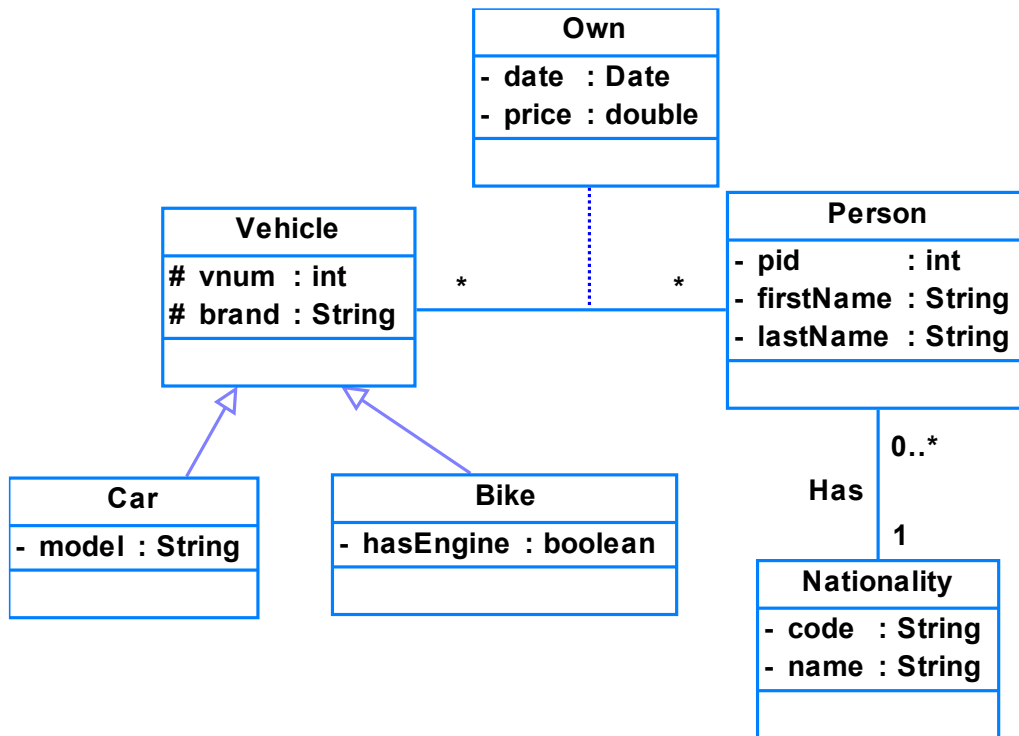
- a- An object oriented language
- b- A procedural language
- c- A logic programming language
- d- An http tag

10- An extent is which of the following?

- a- A keyword that indicates that the subclass inherits from a superclass
- b- A keyword that indicates that the superclass inherits from a subclass
- c- The set of all instances of a class within a database
- d- Only one instance of a class within a database

II- Use ODL language to transform the following class diagram into an object database schema.

(4 pts)



III- A multinational engineering company has decided to distribute its projects information at the regional level. The current centralized relational database schema is as follows:

(6 pts)

Region (rid, rname)  
 Department (did, dname, # rid)  
 Project (pid, pname, pbudget, # did)  
 Employee (eid, name, sex, salary, taxNo, # did)

Departments are grouped regionally as follows:

- Region 1: Middle East
- Region 2: Western Europe
- Region 3: North America

As well as distributing the data regionally, there is an additional requirement to access the employee data either by **personnel information system** or by **payroll information system**.

- a- Produce a distributed database designed for this company.
- b- For each site, define each fragment in relational algebra and specify its type.
- c- Reconstruct the original global relations from their fragments.